

Curriculum Vitae (CV)

VIVEK GUPTA

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Education

2021-Ongoing: Ph.D. - Chemical Engineering, IIT Tirupati

2019-2021: M.Tech- Food Engineering and Technology, Tezpur University

2015-2019: B.Tech- Dairy Technology, Sri Venkateswara Veterinary University

2014: Intermediate, UP Board

2012: 10th, UP Board

National level exams qualified

ICAR ASRB NET – 2023 (Dairy Technology)

GATE- 2020 (Life Sciences)

Publications

(During Ph.D.)

- Gupta, V.,** Saraswat, V., Alagarasan, G., Nallamilli, T., Ramireddy, E., & KSMS, R. (2025). Ultrasound-Assisted Extraction of Betalains from the Amaranthus Using Natural Deep Eutectic Solvents. *Plant Foods for Human Nutrition*, 80(1), 15.
(Impact Factor: 4.0) DOI: <https://doi.org/10.1007/s11130-024-01252-0>
- Saraswat, V., **Gupta, V.,** Alagarasan, G., Nallamilli, T., Ramireddy, E., & Raghavarao, K. S. M. S. (2025). Machine learning-based optimization of integrated extraction method for R-phycoerythrin from dry biomass of Gracilaria corticata. *Algal Research*, 103986.
(Impact Factor: 4.6) DOI: <https://doi.org/10.1016/j.algal.2025.103986>
- Gupta, V.,** Saraswat, V., Nallamilli, T., Nabil, J., & Raghavarao, K. S. M. S. (2025). Machine Learning and RSM-based Optimization of Ultrasound-Assisted Extraction and Purification of Betalains from Amaranthus. *Separation and purification technology*
(Impact Factor: 8.2) (Under Review).
- Saraswat, V., **Gupta, V.,** Alagarasan, G., Nallamilli, T., Ramireddy, E., Raghavarao, KSMS. (2025). Macroalgae (Seaweed) Based Bioactive Compounds with Health Benefits for Functional Foods Development. Book Chapter. (Elsevier, Under process)

(Before Ph.D.)

- Gupta, V.,** Thakur, R., & Das, A. B. (2021). Effect of natural deep eutectic solvents on thermal stability, syneresis, and viscoelastic properties of high amylose starch. *International Journal of Biological Macromolecules*, 187, 575-583
(Impact Factor: 8.025) DOI: <https://doi.org/10.1016/j.ijbiomac.2021.07.099>
- Gupta, V.,** Thakur, R., Barik, M., & Das, A. B. (2022). Effect of high amylose starch-natural deep eutectic solvent-based edible coating on quality parameters of strawberry during storage. *Journal of Agriculture and Food Research*
(Impact Factor: 3.48) DOI: <https://doi.org/10.1016/j.jafr.2022.100487>
- Thakur, R., **Gupta, V.,** & Das, A. B. (2022). Effect of anthocyanin-natural deep eutectic solvent (lactic acid/fructose) on mechanical, thermal, barrier, and pH-sensitive properties of polyvinyl alcohol-based edible films. *Food Packaging and Shelf life*
(Impact Factor: 8.749) DOI: <https://doi.org/10.1016/j.fpsl.2022.100914>
- Thakur, R., **Gupta, V.,** Dhar, P., Deka, S. C., & Das, A. B. (2022). Ultrasound-assisted extraction of anthocyanin from black rice bran using natural deep eutectic solvents: Optimization, diffusivity, and stability. *Journal of Food Processing and Preservation*, 46(3).
(Impact Factor: 2.609) DOI: <https://doi.org/10.1111/jfpp.16309>
- Thakur, R., **Gupta, V.,** Singha, P., & Das, A. B. (2024). Technological and Analytical Aspects of Bioactive Compounds and Nutraceuticals from Plant (Plantation Crops) Sources. In *Bioactive Compounds and Nutraceuticals from Plant Sources* (pp. 43-78). Apple Academic Press. DOI: [10.1201/9781003455172-3](https://doi.org/10.1201/9781003455172-3)

International/National Conferences

(During Ph.D.)

- Gupta, V.,** Saraswat, V., Nallamilli, T., Ramireddy, E., Raghavarao, KSMS. Green Extraction of Betalains from the Amaranthus Employing Field-Assisted Extraction and Natural Deep Eutectic Solvents. *International conference - Food Innovation Asia Conference 2024*. 13-14 June 2024 Bitec, **Bangkok, Thailand** (Honourable Mention Student Graduation Poster Award).
- Gupta, V.,** Saraswat, V., Nallamilli, T., Ramireddy, E., Raghavarao, KSMS. Design and development of an acoustic mist bioreactor for the cultivation of Amaranthus hairy roots. *10th International Conference on Food Chemistry & Technology*. November 25-27, 2024, **Valencia, Spain** (Poster).
- Gupta, V.,** Saraswat, V., Nallamilli T., E., Raghavarao, KSMS. *Extraction and Purification of Betalains from the Amaranthus using Ultrasound-Assisted Extraction followed by Aqueous Two-Phase Extraction*. National Conference on Emerging Technologies for Sustainable Agro-Food-Bio Systems (ET-SAFE-2025). 13-15 February 2025 at **Tezpur University, Assam** (3rd prize in best poster).
- Gupta, V.,** Nallamilli, T., and Raghavarao, KSMS. *Design and development of bioreactor for the cultivation of Amaranthus hairy roots and downstream processing of betalains*. **PMRF Symposium 2025**, 16 to 17th March 2025, **IIT Hyderabad** (Poster).
- Gupta, V., Saraswat, V., **Raghavarao, KSMS**. Extraction and Purification of Bioactive Compounds from Micro/Macroalgae and Hairy Root Cultures. *10th International Conference on Food Chemistry & Technology*. November 25-27, 2024, Valencia, Spain (Invited talk).
- Saraswat, V.,** Gupta, V., Nallamilli, T., Raghavarao, KSMS. Extraction and Purification of R- phycoerythrin from Macroalgae. *10th International Conference on Food Chemistry & Technology*. November 25-27, 2024, Valencia, Spain (Poster).
- Gupta, V., Saraswat, V., **Raghavarao, KSMS**. Designing Bioreactors for the cultivation of Plant cell and Hairy root culture and downstream processing of Bioactive Compounds. ICAR, **New Delhi**, and the Ministry of AFW, 28-30 January 2025, (Invited talk).

Honours and Awards

- Gold Medal - T&I Best Post Graduate Award of Tezpur University from Hon'ble President of India Shri Ram Nath Kovind Ji (2022)
- International Springer Award for Best Paper, *International Conference (Virtual) Sustainable Approaches in Food Engineering and Technology (SAFETY-2021)*
- Association of Food Scientists & Technologists (India) Award for Best Poster (SAFETY-2021)
- 1st prize for flash presentation on Zero Hunger on World Food Day, 16th October 2019, from the Department of Food Engineering and Technology, Tezpur University